

# Nutation Research

*World action models for the hardest movements humans perform.*





Nutation Research

FOUNDER



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Caltech



Massachusetts  
Institute of  
Technology



innovation.  
onto



## THE PROBLEM

# World models for robotics and graphics still struggle with complex biomechanics.

### 1. Most datasets only contain simple demonstrations.

Most world models are only trained on videos demonstrating simple tasks with **no counterfactual data** which tells these models what is wrong and how to improve their motions with reinforcement learning.

### 2. Existing benchmarks don't verify the physics well.

Ask for rotations in the air and the body flails and angular momentum isn't conserved. A model can **break conservation** on every clip and **still lead the field**.

### 3. The field is 62x short on data.

Tencent's 3,000 hour dataset behind HY-Motion 1.0 is **324M** frames. Their one-billion-parameter model should be using roughly **20B** for good generalization. that's **62x more than what exists**. Text models train on 46,296x more.

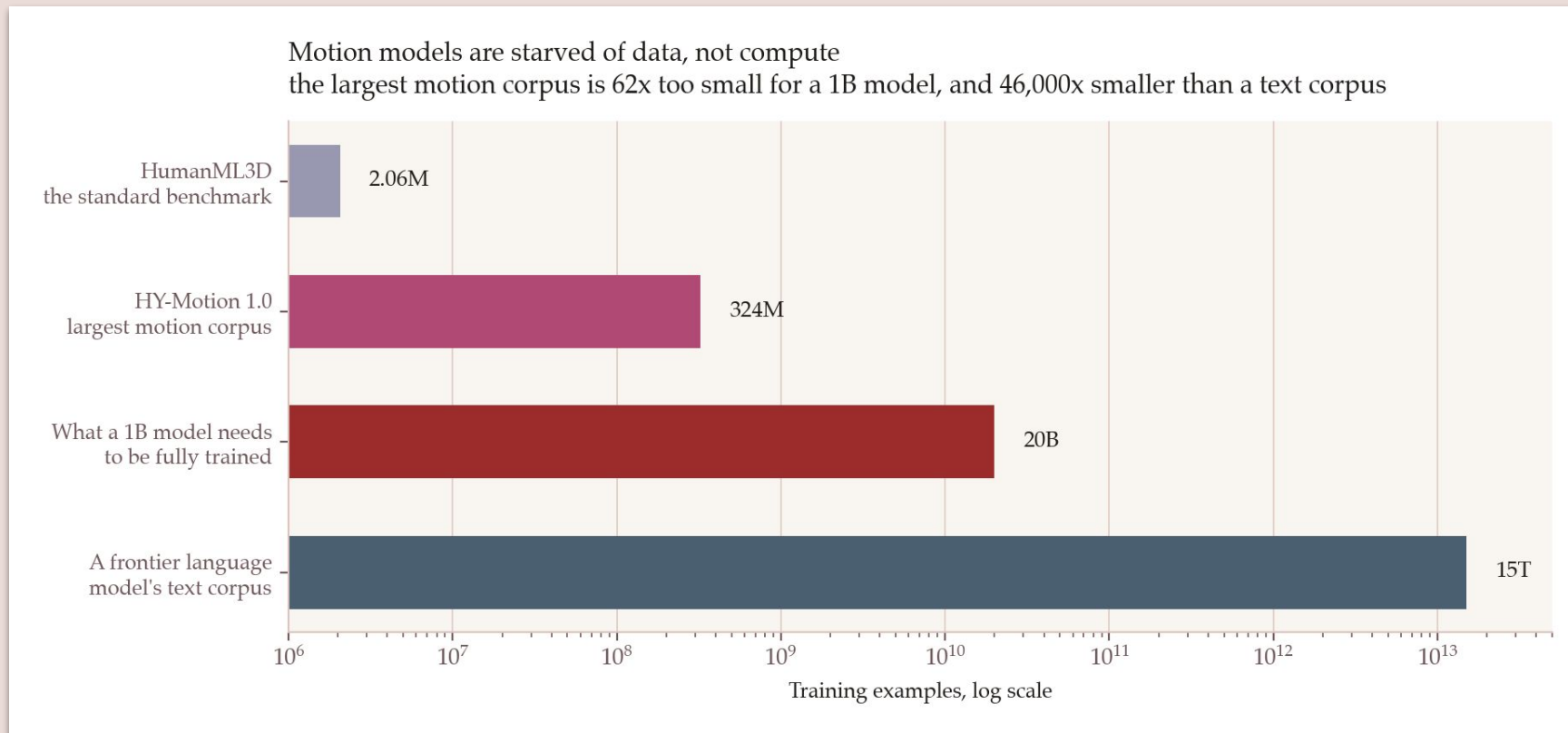
### 4. And the data is the cost, not the compute.

Retraining HY-Motion costs about \$4,582 of GPU. The data behind it costs **72x to 697x** that. Tencent's own scaling work concluded data curation is "the decisive factor."



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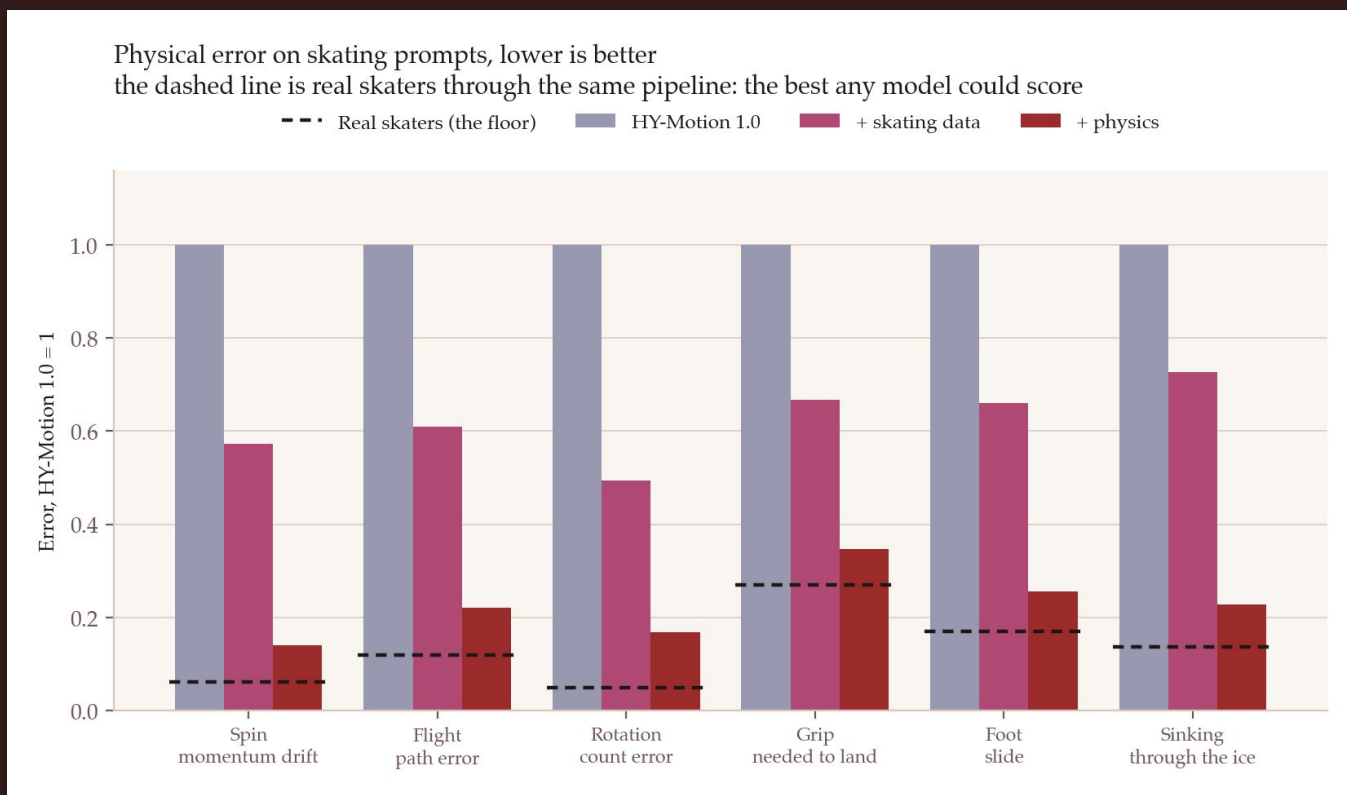
## THE SOLUTION

# Yuna is the instrument to push our world model performance beyond SOTA.

General Intuition built a consumer product, Medal, to collect gameplay data before it rejected \$500M from OpenAI.

We're building the same for the physical space, for the most complex biomechanical motion humans can perform to train world models with RL.

**We collect the data with a tool people already want to use.**



## Generalization

On held-out everyday prompts, such as tripping on a curb, slipping on a wet floor, carrying a box downstairs (lower error is better performance):

- grip needed to stay upright fell **0.61 → 0.34**
- sinking through the floor **19 mm → 8 mm**
- flight path error **9.4 cm → 3.1 cm**.



## YUNA · THE FIRST PRODUCT

# Feed it a skating video. It hands back the better skate.

Yuna watches ordinary broadcast or rinkside footage, detects every jump and spin, grades it, and then shows the athlete the version of that attempt they could have landed, with actionable feedback to get there

### ● Coaching

Per-attempt breakdown plus the specific correction to practice next.

### ● Judging

Accurate scoring with interpretability displayed over the motion trail.

### ● Broadcast

Show the fan the landed version beside the failed one, with the score explained.

### ● → Data

Every interaction writes a labeled, counterfactual-rich example back to the model.

LIVE ELEMENT



4T · GOE +1.5

air time 0.61s · height 58cm · rotation 4.02



## WHY SKATING FIRST

The largest expert-graded motion dataset on earth is unclaimed, and the skating governing body needs the technology now.

### Skating federation is investing in AI.

ISU has been expanding integrations with technology: a January 2026 sensor-data partnership and a February 2026 pilot using AI to support technical scoring. Neither of these use **SOTA world model technology**.

### Figure skating is biomechanically complex.

If a world model can accurately predict and correct a skater landing on a **4mm blade with 5G's of force**, then a robot recovering its balance when it bumps into a kitchen counter is a **trivial sub-problem**.

### Preference data is readily available.

Tencent paid to annotate 40,000 comparison pairs to teach HY-Motion which of two motions was better. Skating has been generating such annotated record (scored against a published rubric) **since 2004**.

### Counterfactual data vastly improves models.

Recent robotics work: a scoring head **trained only on successes made the model worse**: 79% against an 82% baseline and it only reached **92%** once it had seen failures too. Figure skating training data provides vast counterfactuals.

*There is a clear need from the skating federation for SOTA technology. We generate synthetic action labels and futures from the public data, and our private data fills in the counterfactuals that the public can't cover. That proprietary dataset is our moat.*



THE MARKET OPPORTUNITY

The product is a good business. The data is the outsized one.

\$10M

ADDRESSABLE ARR IN FIGURE SKATING

~\$50K-100K per annual contract with clubs,  
teams, federations, and broadcast rights holders.  
~\$1K per individual athlete/coach x ~10K users

\$500M

ADDRESSABLE ARR ACROSS SPORTS

Scale the same model to gymnastics, diving, and  
every other judged or coached discipline.  
Eventually reach sports with multiplayer: MMA,  
American football, etc.

\$2B - 3B

DATA & MODEL CONTRACTS

Robotics, physical AI, and 3D graphics companies  
licensing the corpus and the world model.





THE ASK

# Seeking a **\$500K** SAFE

A small SAFE to secure the first few customers and publish our research before a seed round.

**\$60K**

**Data operation**

Coach/athlete data labeling.

**\$200K**

**Team & ops**

Research, engineering, sales.

**\$80K**

**Compute**

GPUs for pre/post training.

**\$60K**

**GTM & partners**

Athlete ambassadors and federation travel.